

BELLABEAT CASE STUDY

How Can a Wellness Company Play It Smart?

Data Analytics Portfolio Project

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Tools: Python (Pandas) | Microsoft Excel

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EXECUTIVE SUMMARY

This case study analyzes Fitbit smart-device data using **Python (Pandas) and Microsoft Excel** to identify consumer activity and sleep trends that could inform Bellabeat's marketing strategy. After cleaning and combining the datasets, the analysis included **1,373 daily activity records from 35 users** and **410 sleep records from 24 users**.

Key findings showed that **80% of users were classified as low or moderately active**, light activity accounted for substantially more recorded active time than higher-intensity activity, and daily steps had a **moderate positive correlation with calories burned ($r = 0.58$)**. Activity and sleep behavior also varied across the week.

Based on these findings, the analysis recommends that Bellabeat focus on **personalized and achievable movement goals, day-specific user engagement, and integrated activity and sleep insights**. Because the dataset consists of Fitbit users rather than Bellabeat customers, these recommendations should be validated using Bellabeat's own customer data before major implementation.

1. ASK

Business Task

The objective of this analysis is to examine smart-device usage data to identify trends in consumers' activity and sleep behavior and determine how these insights could help Bellabeat improve its marketing strategy and user engagement.

Key Business Questions

1. What are some trends in smart-device usage?
2. How could these trends apply to Bellabeat customers?
3. How could these trends help influence Bellabeat's marketing strategy?

Stakeholders

The primary stakeholders are Bellabeat cofounders Urška Sršen and Sando Mur, along with the Bellabeat marketing analytics team.

2. PREPARE

Data Source

For the case study, we used the **Fitbit Fitness Tracker Data** provided through Kaggle for the Bellabeat case study. The dataset contains personal fitness tracker information such as daily activity, steps, calories, activity intensity, and sleep.

Although the full dataset contains multiple files at different levels of detail, we selected the datasets most relevant to our business questions.

Datasets Used

Two datasets were selected for the core analysis:

1. `dailyActivity_merged.csv`

This file contained daily activity measures including:

- Total steps
- Total distance
- Very active minutes
- Fairly active minutes
- Lightly active minutes
- Sedentary minutes
- Calories

We combined daily activity data from two available periods:

March 12, 2016 – April 12, 2016

April 12, 2016 – May 12, 2016

After resolving the overlapping date, the final activity dataset contained:

1,373 daily records from 35 unique users.

2. `sleepDay_merged.csv`

This file contained sleep measures including:

- Total minutes asleep
- Total time in bed
- Number of sleep records

After removing duplicates, the sleep dataset contained:

410 daily sleep records from 24 unique users, covering April 12 – May 12, 2016.

Why We Selected These Data

These datasets allowed us to investigate three areas relevant to Bellabeat:

Daily activity → Activity behavior and weekday patterns

Calories → Relationship between movement and energy expenditure

Sleep → Sleep duration and weekly patterns

This kept our analysis focused on the business problem instead of analyzing every file simply because it was available.

Data Limitations

The dataset has several limitations. It represents a relatively small sample of Fitbit users and contains data collected in 2016, so the results may not represent current smart-device behavior or Bellabeat's customer base. User participation also varied, with some users contributing considerably fewer days of activity data than others. Sleep data was available for only 24 of the 35 activity users. Therefore, the findings should be treated as exploratory insights rather than conclusions about all Bellabeat customers.

3. PROCESS

Tools Used

Python (Pandas) in Kaggle Notebook was used for data loading, inspection, cleaning, transformation, merging, and analysis. **Microsoft Excel** was used to create and format the final visualizations.

Data Inspection

The datasets were first inspected for their dimensions, column names, data types, missing values, duplicate records, date ranges, and number of unique users. This helped identify data-quality issues before beginning the analysis.

During inspection, we found:

- The two daily activity files contained **457 and 940 records** respectively.
- There were **no missing values** in the selected activity or sleep datasets.
- The combined activity data contained overlapping records for **April 12, 2016**.
- The sleep dataset initially contained **413 records**, including **3 exact duplicate rows**.

Data Cleaning

The two daily activity datasets were combined to cover the full analysis period. After combining them, 24 duplicate user-date records were identified on April 12, 2016, where the two collection periods overlapped. The duplicate records from the earlier period were removed while retaining the corresponding records from the later period. This resulted in **1,373 cleaned daily activity records from 35 users**.

The sleep dataset contained three exact duplicate rows, which were removed, leaving **410 sleep records from 24 users**. Date columns were converted to a consistent datetime format to support date-based analysis and merging.

Zero-step records were reviewed separately as part of the cleaning process:

The activity dataset contained **133 zero-step records (9.69%)**. These records were retained rather than automatically removed because a zero-step day could represent either genuine inactivity or incomplete device usage. An `ActiveDay` indicator was created to distinguish records with recorded steps from zero-step records without altering the original observations.

Data Transformation

A `DayOfWeek` variable was created from the activity date to analyze weekly activity patterns. Average daily steps were also calculated for each user, and users were grouped into project-defined activity segments: **Low Activity (<5,000 steps)**, **Moderate Activity (5,000–9,999 steps)**, and **High Activity (≥10,000 steps)**. These categories were created specifically for this analysis and should not be interpreted as official Bellabeat or Fitbit classifications.

For sleep analysis, two additional measures were derived:

Sleep duration was converted into hours for easier interpretation, and sleep efficiency was calculated by comparing total minutes asleep with total time in bed.

Activity and Sleep Merge

The cleaned activity and sleep datasets were joined using both **user ID and date**. An inner join was used so that only dates with both activity and sleep information were included in the combined analysis. The resulting dataset contained **410 matched daily records from 24 users**.

4. ANALYZE

Daily Activity Patterns

Users averaged approximately **7,377 steps per day** across the cleaned activity dataset. Activity also varied by day of the week. **Saturday recorded the highest average at 7,752 steps**, while **Sunday recorded the lowest at 6,607 steps**.

User Activity Segmentation

Average daily steps were calculated for each of the 35 users and used to create three project-defined activity segments. **48.6% of users were classified as moderately active, 31.4% as low activity, and 20.0% as high activity**. Therefore, **80% of users fell within the low or moderate activity groups** under the thresholds used in this analysis.

Activity Intensity

Users spent considerably more time performing light activity than higher-intensity activity. Average daily activity consisted of approximately **188.1 lightly active minutes, 19.9 very active minutes, and 13.6 fairly active minutes**.

Steps and Calories

The relationship between daily steps and calories burned was examined using correlation analysis. A **moderate positive correlation ($r = 0.58$)** was identified, indicating that higher-step days generally tended to be associated with higher calorie expenditure.

Sleep Patterns

Sleep analysis was limited to the **24 users with available sleep records**. These users averaged approximately **6.99 hours of sleep per night**. Sleep duration also varied across the week, with **Sunday recording the highest average at 7.55 hours and Thursday the lowest at 6.69 hours**.

Activity and Sleep Relationship

Additional correlation analysis did not identify a meaningful positive relationship between sleep duration and either daily steps or calories burned. These results were therefore treated as supporting analysis rather than primary findings.

5. SHARE

The following visualizations highlight the key trends identified from the cleaned Fitbit activity and sleep data.

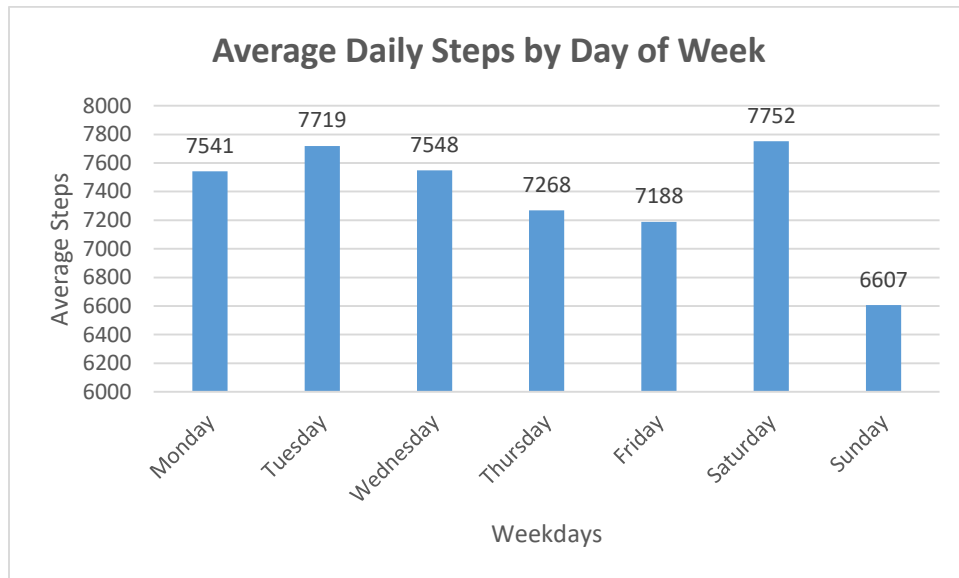


Figure 1: Average Daily Steps by Day of Week

Average daily steps varied across the week. Saturday recorded the highest average activity at **7,752 steps**, while Sunday recorded the lowest at **6,607 steps**. This suggests that the timing of activity engagement may be relevant when designing personalized reminders or challenges.

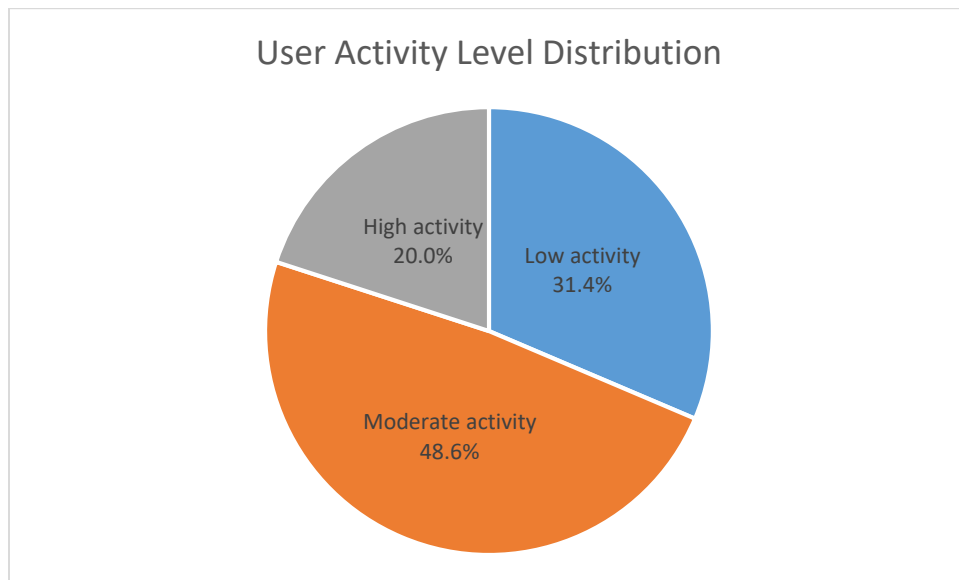


Figure 2: User Activity Level Distribution

Based on the activity segments defined for this analysis, **48.6% of users were moderately active, 31.4% were low activity, and 20.0% were highly active.** Overall, **80% of users fell into the low or moderate activity groups**, suggesting that achievable everyday movement goals may be relevant to a large portion of users.

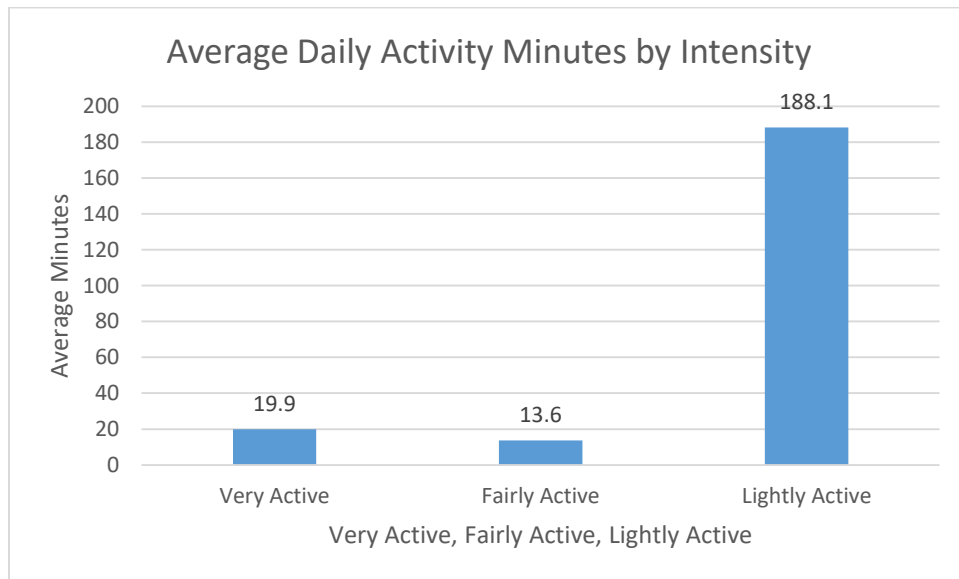


Figure 3: Average Daily Activity Minutes by Intensity

Light activity accounted for substantially more recorded active time, averaging **188.1 minutes per day**, compared with **19.9 very-active minutes** and **13.6 fairly-active minutes**. This indicates that everyday light movement represents a significant part of users' recorded activity and could be an important area for Bellabeat's user-engagement strategy.

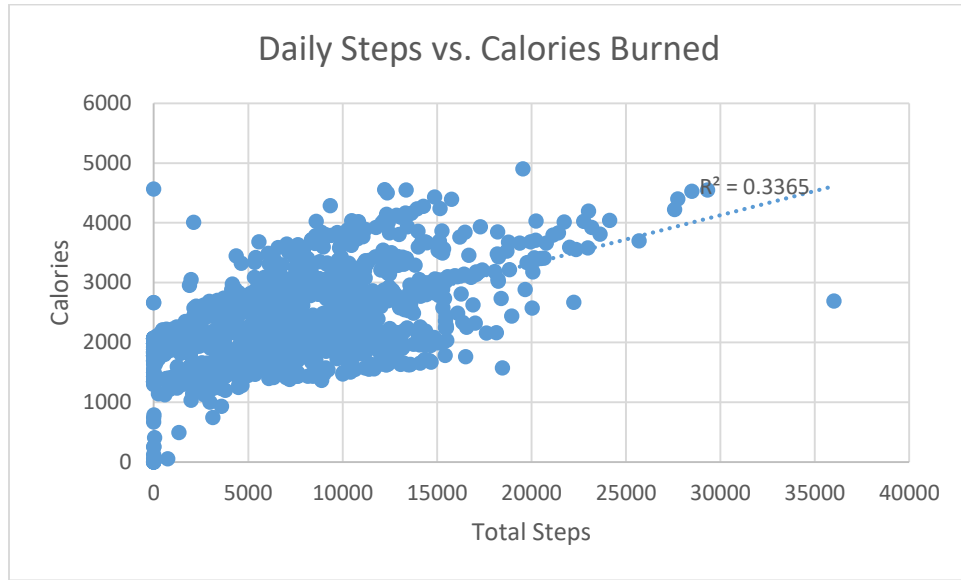


Figure 4: Relationship Between Daily Steps and Calories Burned

Daily steps showed a **moderate positive correlation with calories burned ($r = 0.58$)**. The linear trendline produced an **R^2 of 0.3365**, indicating that approximately **33.7% of the variation in calories burned was explained by daily steps within this simple linear model**. Overall, higher-step days tended to be associated with greater calorie expenditure, although other factors also contribute to calories burned.

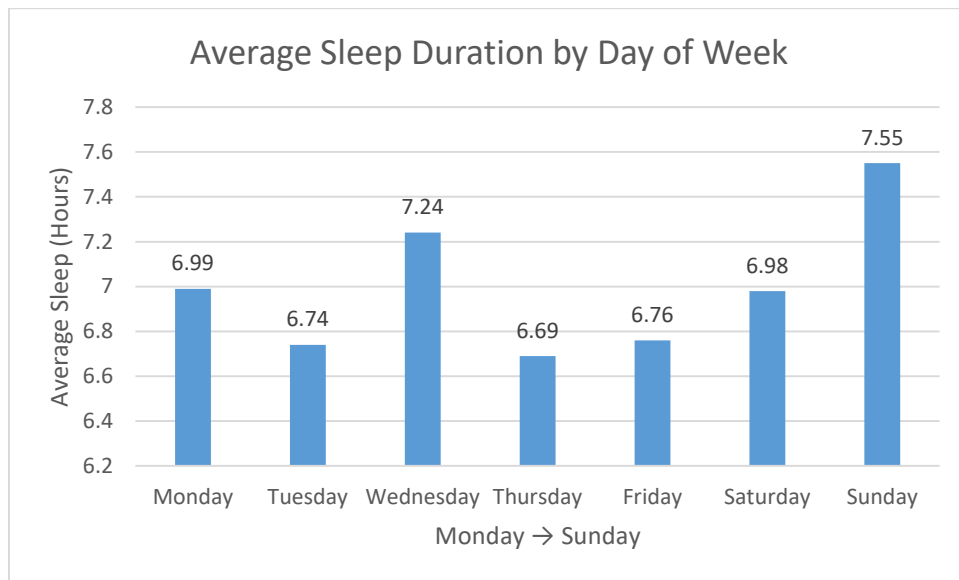


Figure 5: Average Sleep Duration by Day of Week

Among the **24 users with available sleep data**, average sleep duration was approximately **6.99 hours per night**. Sleep duration varied across the week, with **Sunday recording the highest average at 7.55 hours** and **Thursday the lowest at 6.69 hours**. These patterns suggest that sleep behavior may provide an additional opportunity for Bellabeat to support users through personalized wellness insights.

6. ACT

The analysis supports three practical recommendations for Bellabeat:

Recommendation 1 — Personalize Achievable Movement Goals

Bellabeat should use its app to provide **personalized and achievable daily movement goals** based on users' activity levels. Since **80% of users in this analysis were classified as low or moderately active** and light activity averaged **188.1 minutes per day**, Bellabeat could emphasize walking, short movement breaks, and gradual step increases rather than focusing primarily on intense workouts.

Recommendation 2 — Use Day-Specific Engagement

Bellabeat could use **personalized reminders, challenges, and motivational messages based on day-of-week activity patterns**. Average steps were highest on Saturday (7,752) and lowest on Sunday (6,607), suggesting that the timing of engagement may be useful when encouraging consistent activity. These interventions should ideally adapt to each user's own behavior rather than assuming the same weekday pattern applies to everyone.

Recommendation 3 — Integrate Activity and Sleep Insights

Bellabeat could combine **activity and sleep information** within its app to provide users with a more holistic view of their wellness. Among users with available sleep data, average sleep was **6.99 hours per night** and varied across the week. Personalized weekly summaries could help users understand their movement and sleep patterns together and support healthier daily routines.

These recommendations are based on exploratory Fitbit data rather than Bellabeat customer data. Bellabeat should validate these patterns using a larger and more representative sample of its own users before implementing major marketing decisions.

7. CONCLUSION

This analysis explored Fitbit smart-device data to identify patterns in daily activity, activity intensity, calorie expenditure, and sleep behavior that could inform Bellabeat's marketing strategy.

The analysis found that most users fell within the low or moderate activity segments, light activity represented the largest portion of recorded active time, and daily steps had a moderate positive relationship with calories burned. Activity and sleep patterns also varied across the week.

Based on these findings, Bellabeat could focus on **personalized movement goals, day-specific engagement, and integrated activity and sleep insights** to support user engagement and promote everyday wellness.

However, because the dataset contains a relatively small sample of Fitbit users and does not represent Bellabeat's own customer base, these findings should be considered exploratory. Future analysis using larger, more recent Bellabeat customer data would help validate these patterns and support more targeted marketing decisions.